

SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA1205	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	C04	Assessment:	ASSESSMENT TOOL3- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
C04 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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Department:	B TECH.IT	Date:	

ANNEXURE A COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

Feature	L1 Cache	L2 Cache	L3 Cache
Latency	Very low, fastest	Low to mod	Higher than
Bandwidth	Very high	High	Moderate to
Throughput	Highest	High	Lower than
Size	Smallest	Medium	Largest
Location	Inside CPU core	Usually near core	Shared cores
Purpose	Frequently data/instructions	Backup for L1	Shared back L1/L2

2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

Feature	SRAM	DRAM
Latency	Very low	Higher
Bandwidth	Very high	High
Throughput	High	Moderate to high
Storage	Flip-flops	Capacitors
Refresh	Not required	Required
Cost	Expensive	Cheaper
Use	Cache memory	Main memory

3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

Feature	Virtual Memory & Paging	Cache Memory
Purpose	Extends memory using secondary storage	Speeds up CPU memory access
Latency	Very high during page faults	Very low
Bandwidth	Lower	Very high
Throughput	Reduced when page faults occur	Increased due to fast access
Storage	SSD/HDD + RAM	SRAM
Management	OS/MMU	Hardware

4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

Feature	NAND Flash	DRAM
Latency	Higher	Much lower
Bandwidth	Moderate to high	High
Throughput	High for sequential access	Very high
Volatility	Non-volatile	Volatile
Write/Erase	Slower	Faster
Use	SSD, storage	Main memory

5.Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Feature	MRAM	RRAM
Access Time	Very fast	Fast
Throughput	High	High
Energy Efficiency	High	Very high
Volatility	Non-volatile	Non-volatile
Endurance	Very high	High
Main Advantage	Fast and reliable	High density and low power

Code of Conduct:

*I **SREYA.B(192521442)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand